

MethylPipeline User Manual

Operational runbook for package usage and staged workflows

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1 MethylPipeline User Manual

1.1 What this manual is

This is the operational companion to the theory book. It is designed for execution teams who need:

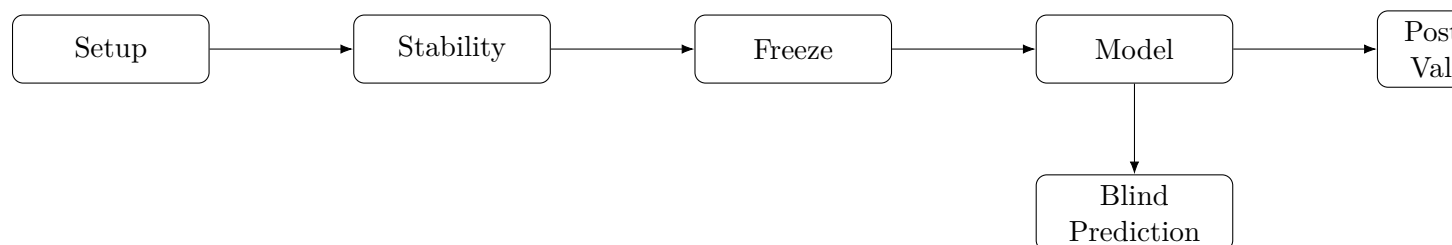
- exact commands,
- required config keys,
- expected output artifacts,
- stage handoff checks,
- recovery actions when a run fails.

For statistical background, use `docs/theory/`.

1.2 Reading paths

- **Sample prep (FASTQ → HDF5):** Chapter 03 (`03-sample-prep-and-qc.qmd`).
- **Fast path (stage execution):** Chapters 05 to 09 (stability through blind prediction).
- **New project setup:** Chapters 01, 02, 03, then 05.
- **Daily operations:** Chapters 10 to 12.
- **Scaling MC across machines / optional hyperparameter tuning:** Chapters 13 and 14 (shared `/work` storage, queue subcommands, and the `methyl-hyperparam-search` grid driver).
- **Standalone package usage:** Chapter 04 (`03-package-reference-individual-usage.qmd`).

1.3 End-to-end workflow map



1.4 Scope boundaries

- **SamplePrepPipeline** (FASTQ → aligned BAM → methylation HDF5) runs before staged validation; see Chapter 3.
- **methvl-validation** orchestrates stability, freeze, model, and post-model validation.
- Blind-only prediction is documented as a standalone **methvl-predictor** operation.
- Package behavior is documented from implementation in **packages/***.

1.5 Canonical references

- Sample prep workflow: [workflow_engine/sql/SamplePrepFlow.md](#)
- Workflow reference: [docs/WORKFLOW_DIAGRAM_PACK_Healthy_vs_PCa1-4-CG.md](#)
- Validation orchestration: [packages/methylvalidation/docs/USAGE.md](#)
- Distributed queue (shared storage, plan-runs / run-task / aggregate-results): [packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md](#)
- Hyperparameter search (objective (J), methyl-hyperparam-search): [packages/methylvalidation/docs/HYPERPARAM_SEARCH.md](#)
- Validation flags and config model:
 - [packages/methylvalidation/methyl_validation/cli.py](#)
 - [packages/methylvalidation/methyl_validation/config.py](#)

Part I

Part I: Setup and Package Orientation

2 Prerequisites and Environment

2.1 Purpose

Set up a reproducible execution environment for all package CLIs and workflow stages.

2.2 Required tools

- Python environment in repo-local `.venv`
- Installed package CLIs (`methyl-*`)
- Quarto for manual rendering (optional for pipeline runs)
- TeX engine for PDF rendering (`quarto install tinytex` recommended)

2.3 Environment bootstrap

From repo root:

```
python3.12 -m venv .venv
source .venv/bin/activate
bash scripts/install_all.sh
```

Optional dependency sets:

```
bash scripts/install_all.sh --pipeline-reqs
bash scripts/install_all.sh --gpu-reqs
```

2.4 Verify CLI availability

```
source .venv/bin/activate
methyl-validation --help
methyl-centroid --help
methyl-detector --help
methyl-mapper --help
methyl-enricher --help
methyl-predictor --help
```

2.5 Runtime assumptions

- Activate `.venv` before every run.
- Use absolute paths in project configs where possible.
- Keep `samples_base_path` consistent across stages.

2.6 Success checks

- `.venv` activates without import errors.
- `methyl-validation --help` prints options.
- Project JSON path resolves on the execution host.

2.7 Common failures and fixes

- **command not found:** activate `.venv` and reinstall with `scripts/install_all.sh`.
- **module import errors:** reinstall editable packages from repo root.
- **path mismatch between hosts:** set `step_config.validation.path_remap` or `--path-remap OLD=NEW`.

2.8 See also

- `README.md`
- `docs/DEPLOYMENT.md`

3 Project Config and Layout

3.1 Purpose

Define a project JSON that can run through stability, freeze, model, and evaluation without per-stage schema drift.

3.2 Repository vs /work (what goes where)

Repository (versioned, documented)	/work/<disease>/ (study-specific)
Pipeline profiles (workflow_engine/domain/profiles/)	project_*.json — cohorts, comparisons, sample paths, output_base, study step_config
DomainPrograms (*.program.json under workflow_engine/domain/)	Sample list CSVs under data/
JSON schemas (schemas/config/)	Run outputs (monte_carlo_runs/ stability/, ...)
User manual, theory, docs/domain_program_language.md	

- Point methyl-workflow-run --program at a **repo** DomainProgram path.
- Point --context / projectPath at a **project JSON** on **/work**.
- Optional --context-file loads a **profile** from workflow_engine/domain/profiles/.
- Do **not** treat repo workflow_engine/domain/checks/*/configs/project_*.json as production configs (CI smoke only).
- Do **not** copy *.program.json into /work/<disease>/configs/.

3.3 Minimum structure

Use one project file containing:

- global cohort metadata (controls, diseases, comparisons),
- filesystem anchors (output_base, samples_base_path),
- step-specific overrides under step_config.

Core stage driver:

- step_config.validation (parsed into MonteCarloConfig by methyl-validation).

Upstream sample prep (before stability) uses:

- `step_config.alignment_qc` — guardrails, cycle screening, optional cfDNA fragmentomics ([Chapter 3](#)).
- `step_config.extraction_qc` — post-extract coverage and conversion thresholds ([Chapter 3](#)).

3.4 Validation stage keys that matter most

- split policy: `train_fraction`, `n_iterations`, `seed`
- stability controls:
 - `run_stability`
 - `stability_dmp_freq`
 - `stability_min_balanced_accuracy`
 - `stability_gene_freq`
 - `stability_featurecuts_enabled`
 - `stability_target_balanced_accuracy`
 - `stability_min_selected_dmps`
- freeze/model paths:
 - `freeze_stable_dmp_csv` (optional)
 - `production_output_dir` (optional)
- backend/model:
 - `model_backend`
 - `feature_mode`
 - `ecdf_second_stage_enabled`
 - backend-specific `tabular_*` / `generative_*` keys

3.5 Output layout

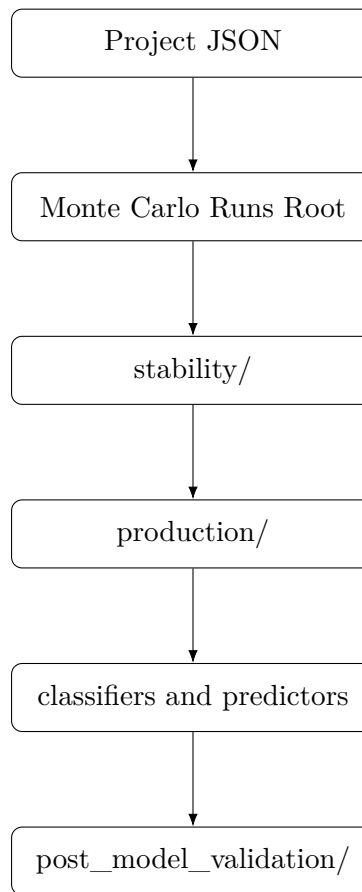
Most runs write to:

`<output_base>/<project_name>/monte_carlo_runs/`

Important subtrees:

- `run_XXXX/` (per-iteration artifacts)
- `stability/` (stable panel and summaries)
- `production/` (freeze and final model outputs)
- `model_mc/` (backend comparison runs)
- `post_model_validation/` (frozen-model descriptive evaluation)

3.6 Handoff model



3.7 Do not do this

- Do not put blind-only predictor config (`predictor.blind`) into validation-driven runs.
- Do not change `samples_base_path` between stages without explicit remap.
- Do not run `--freeze` before stable panel exists.

3.8 Success checks

- Project loads in `methvl-validation --project ....`
- `step_config.validation` exists and parses.
- Run directories appear under expected root.

3.9 See also

- `docs/theory/chapters/11-project-configuration.qmd`
- `docs/theory/chapters/13-configuration-reference.qmd`

4 Sample Prep and Quality Control

4.1 Purpose

SamplePrepPipeline is the per-sample upstream workflow that turns raw WGBS FASTQs into methylation HDF5 files (`{chromosome}-CG.h5`) that downstream stages consume. Every sample passes through **two automated guardrails** before it enters stability, freeze, or discovery workflows:

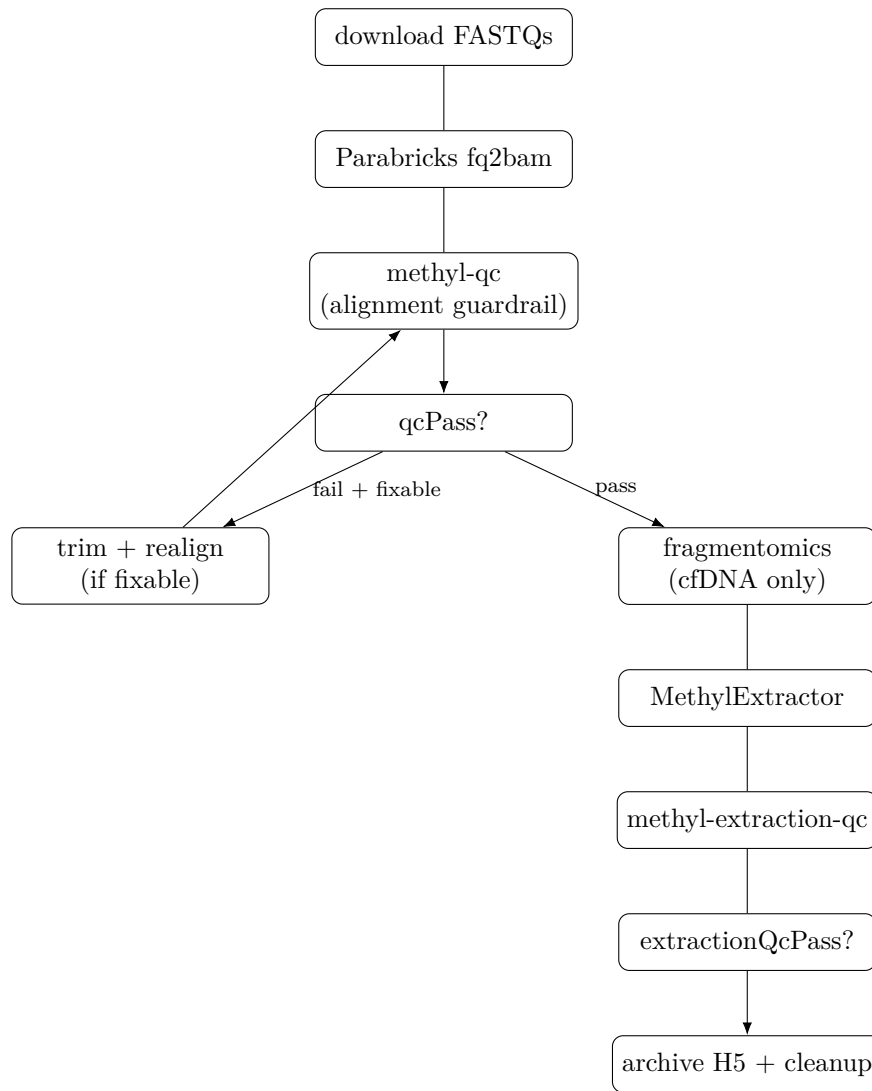
1. **Alignment QC** (`methyl-qc`) — confirms Parabricks alignment and sequencing metrics are acceptable (or fixable via trim/realign).
2. **Extraction QC** (`methyl-extraction-qc`) — confirms MethylExtractor produced sufficient coverage and conversion-quality signal across chromosomes.

Without these gates, bad alignments would waste GPU extraction time, and under-covered or poorly converted extractions would pollute centroids, DMP discovery, and classifiers.

Workflow source of truth: [workflow_engine/domain/fixtures/sample_prep.program.json](#)

Operator reference: [workflow_engine/sql/SamplePrepFlow.md](#)

4.2 End-to-end flow



Gate	Scope variable	When	Blocks
Alignment	<code>qcPass</code>	After Parabricks, before extract	Extraction (unless remediation succeeds)
Extraction	<code>extractionQcPass</code>	After MethylExtractor	Full archive and BAM deletion

Ordering rationale: alignment QC runs before cfDNA fragmentomics so failed samples skip BAM scanning. Fragmentomics and extraction both need the aligned BAM. FASTQs are retained until QC is fully resolved so **fastp** can trim and Parabricks can realign on the remediation path.

4.3 Where artifacts live

Per sample under `/work/samples/{sample_id}/`:

Artifact	Role
<code>*deduplicate_metrics.txt</code>	Picard-style duplication metrics from Parabricks
<code>{sample_id}.json</code>	Parabricks WGBS metrics JSON (sequencing + alignment)
<code>{sample_id}.qc-metrics.tar</code>	Optional tar of tabular metrics (used when JSON is absent)
<code>{output_base}/{project}/alignment_qc/{sample_id}.json</code>	Normalized V2 alignment QC export (when run project-scoped)
<code>{sample_id}.extraction_manifest.json</code>	MethylExtractor summary (written at extract time)
<code>{sample_id}.extraction_qc.json</code>	Extraction guardrail report
<code>{sample_id}.sample_prep_log.jsonl</code>	Append-only audit of every prep action
<code>{chr}-CG.h5</code>	Methylation matrices consumed by <code>methyl-centroid</code>

4.4 Alignment QC: inputs and normalized JSON

`methyl-qc` (`packages/methylalignmentqc`) combines **two upstream metric families** into one decision:

4.4.1 Picard-style duplication metrics

Parabricks emits a tab-separated **deduplication metrics** file (names vary: `*deduplicate_metrics.txt` or `*duplication_metrics.txt`). The parser (`methyl_alignment_qc/core/parser.py`) reads Picard-compatible columns such as:

- `LIBRARY`, `UNPAIRED_READS_EXAMINED`, `READ_PAIRS_EXAMINED`
- `SECONDARY_OR_SUPPLEMENTARY_RDS`, `UNMAPPED_READS`
- `UNPAIRED_READ_DUPLICATES`, `READ_PAIR_DUPLICATES`
- `PERCENT_DUPLICATION`

These feed `summary_stats` on the export JSON (total reads, duplication rate, etc.). Optional config guardrails (`step_config.alignment_qc.optional_guardrails.duplication_rate_max`) can fail samples with excessive PCR duplication even when cycle quality looks fine.

4.4.2 Parabricks WGBS metrics JSON

Each sample directory should contain `{sample_id}.json`: Parabricks’ structured report with sections such as:

- `quality_yield` — total/PF reads and Q30 bases
- `mean_quality_by_cycle` — per-cycle mean Phred (Read 1 then Read 2; for 151+151 cycles, Read 2 starts at cycle 152)
- `gc_bias_summary, insert_size_metrics`
- `pre_adapter_summaries` — deamination and OxoG artifact qscores

When this file is present, guardrails in `wgbs_parabricks_qc.py` evaluate sequencing and alignment health **before** methylation extraction.

4.4.3 Our equivalent export JSON (V2)

The worker writes a normalized **V2 row-oriented** JSON per sample (schema: `schemas/config/alignment_qc/exp`

Key blocks:

Block	Contents
<code>summary_stats</code>	Duplication rate, read counts from Picard metrics
<code>quality_yield,</code> <code>mean_quality_by_cycle, ...</code> <code>guardrails.details</code>	Parabricks sections (embedded or referenced) Per-metric pass/fail, observed value, normal range, operator message
<code>guardrails.screening</code> <code>guardrails.overall_pass</code> <code>qc_history</code>	Cycle-quality disposition and recommended trim counts Single boolean bound to workflow <code>qcPass</code> Append-only list of evaluations (initial + post-remediation retries)
<code>fragmentomics_metrics</code>	cfDNA insert-size guardrails when enabled

V1-shaped payloads are assembled internally and converted to V2 for export; standalone tools can migrate older files with `methy1-qc-migrate-guardrails`.

4.5 Alignment guardrails: what “good alignment” means

Core WGBS guardrails (defaults in `wgbs_parabricks_qc.py`) include:

Metric	Default expectation	Why it matters
PF read percent	90%	Run-level filtering quality
Q30 percent	85% (configurable)	Base-call confidence
Mean quality	35	Overall sequencing noise
Min quality after cycle 20	30	Late-cycle degradation
AT / GC dropout	< 3 / < 5	Coverage bias across genome

Metric	Default expectation	Why it matters
Median insert size	150–300 bp	Library fragmentation sanity
Deamination qscore	30	Expected WGBS conversion signal
OxoG qscore	20	Oxidative damage risk

Each metric carries a **message** explaining the biological/operational concern. `guardrails.overall_pass` is the logical AND of all evaluated checks (including optional and analyte-specific add-ons).

4.5.1 Optional and analyte-specific checks

Configure under `step_config.alignment_qc` in the project JSON:

```
{
  "alignment_qc": {
    "optional_guardrails": {
      "duplication_rate_max": 0.25,
      "min_pf_reads": 1000000
    },
    "bisulfite_conversion": {
      "enabled": true,
      "source": "auto",
      "min_conversion_rate_pct": 99.0
    },
    "fragmentomics": {
      "enabled": true,
      "profile": "cfDNA"
    },
    "cycle_screening": {
      "enabled": true,
      "r2_quality_threshold": 30,
      "max_trim_bases": 8
    }
  }
}
```

- **Bisulfite conversion:** reads `bisulfite_conversion.json` sidecar when present; otherwise may use deamination qscore as a qualitative proxy.
- **cfDNA fragmentomics:** when `primary_analyte` is `cfDNA`, insert-size histogram guardrails can flag abnormal fragment profiles before extraction.

See [docs/ANALYTE_PROFILES.md](#) for profile defaults.

4.6 Cycle screening and automated recovery

Historically, a single `min(mq[20:])` over the combined cycle series could fail fixable Read-2 start dips with a generic “do not proceed” message. **Cycle screening** (`cycle_quality_screening.py`) inspects Read 1 and Read 2 separately and assigns a **disposition**:

Disposition	Meaning	SamplePrep action
USE_CURRENT_ALIGNMENT	Guardrails pass	Continue to extract
REALIGN_TRIM / REALIGN_READ2_TRIM	Fixable read-end quality dip	<code>sample.trim_fastq</code> → Parabricks <code>forceRealign</code> → methyl-qc retry
INVESTIGATE_MULTI_REGION	Multiple low-quality regions	Manual review; no auto-trim
INVESTIGATE_GUARDRAIL_ONLY	Cycles OK; duplica- tion/depth/insert fail	Manual review
NOT_FIXABLE	No automated path	Archive QC-only artifacts, mark failed

Screening writes `guardrails.screening` with recommended trim bases (`trim_front1`, `trim_tail1`, `trim_front2`, `trim_tail2`). SamplePrep binds `remediateAlignment` when disposition is trim-eligible and FASTQs are still on disk.

Important: FASTQs must **not** be deleted before final alignment QC disposition. The workflow deletes FASTQs only after pass or final reject (including post-remediation retry).

Offline cohort review (no re-align):

```
source .venv/bin/activate
python scripts/alignment_qc_cohort_screening.py \
  --qc-dir /work/prostate-cancer/alignment_qc \
  --group healthy=/work/prostate-cancer/data/healthy.csv \
  --group pca=/work/prostate-cancer/data/pca.csv \
  --out /work/prostate-cancer/alignment_qc/screening_report
```

4.7 Extraction QC: confirming enough good reads

Alignment QC validates the **BAM**. Extraction QC validates the **methylation call layer** MethylExtractor produced.

4.7.1 Input: extraction manifest

MethylExtractor writes `{sample_id}.extraction_manifest.json` with:

- `summary` — genome-wide CpG weighted mean coverage, CHH/CHG methylation levels
- `per_chromosome` — per-chromosome blocks with `CG.mean_coverage` and related stats

4.7.2 Output: extraction QC JSON

`methyl-extraction-qc` reads the manifest and writes `{sample_id}.extraction_qc.json`. Default guardrails (`methylextractionqc/guardrails.py`):

Guardrail	Default	Interpretation
<code>cpg_weighted_mean_coverage</code>	10×	Enough reads at CpG sites passing minimum coverage
<code>chh_methylation_level</code>	0.02	Low non-CpG methylation (conversion sanity) when CHH extracted
<code>chg_methylation_level</code>	0.02	Same for CHG
<code>chromosome_completeness</code>	all expected chromosomes	Every project chromosome present with CG metrics
<code>chromosome_uniformity</code>	autosomal min/median 0.5	No severe localized dropout

`guardrails.overall_pass` maps to workflow `extractionQcPass`. Failures are **terminal** today (no automatic retry loop).

Override thresholds via `step_config.extraction_qc` in the project JSON. Schema: `schemas/extraction_qc_output.json`

4.7.3 Why both gates matter

Question	Answered by
Did sequencing and alignment produce a usable BAM?	Alignment QC
Did we extract enough confident methylation signal genome-wide?	Extraction QC

A sample can pass alignment but fail extraction (e.g., coverage collapse on certain chromosomes). Conversely, remediation after alignment QC prevents spending extract GPU time on BAMs that will never support stable downstream statistics.

4.8 Pass, fail, and handoff

Outcome	Alignment QC	Extraction QC	Next step
Pass both	<code>qcPass: true</code>	<code>extractionQcPass: true</code>	Optional full archive → delete FASTQs/BAM → sample ready for DataDrivenPipeline

Outcome	Alignment QC	Extraction QC	Next step
Fail alignment (not fixable)	<code>qcPass: false</code>	—	QC-only archive, mark failed, skip extract
Fail after remediation	<code>qcPass: false</code> (attempt 2)	—	Same as above
Pass alignment, fail extraction	<code>qcPass: true</code>	<code>extractionQcPass: false</code>	QC-only archive, mark failed; H5 may exist but must not enter production cohorts

When SamplePrep instance **COMPLETED**, start stability or discovery workflows with the same `projectPath`. Downstream code resolves HDF5 via `methyl_domain.helpers.resolve_methylation_h5_path()` — do not hard-code sample paths.

Workers tag scope with `MethylSampleRef` including `alignmentQc`, `extractionQc`, and `methylation` refs for traceability.

4.9 Configuration checklist

Before starting SamplePrep, confirm:

- ☐ `project.json` lists samples and `samples_base_path` / `path_remap` resolve to `/work/samples/...`
- ☐ `step_config.alignment_qc` thresholds match analyte (cfDNA vs buffy coat)
- ☐ `step_config.extraction_qc` min coverage appropriate for WGBS depth expectations
- ☐ `validation.regulatory.primary_analyte` set (drives fragmentomics profile)
- ☐ Instance `context_json` includes `fastqStorage`, `referenceFasta`, `samples[]`

4.10 Commands (standalone debugging)

```
source .venv/bin/activate

# Alignment QC for one sample directory
methyl-qc --samples /work/samples/SAMPLE_ID --output-dir /work/prostate-cancer/alignment_qc

# Project-scoped (uses step_config.alignment_qc)
methyl-qc --project /work/prostate-cancer/configs/project_Example.json

# Extraction QC after MethylExtractor
methyl-extraction-qc --sample-dir /work/samples/SAMPLE_ID --sample-id SAMPLE_ID
```

4.11 Related documentation

- Workflow operator guide: [workflow_engine/sql/SamplePrepFlow.md](#)
- Architecture §0: [workflow_engine/docs/pipeline_architecture.md](#)
- Worker contract: [workflow_engine/contract/sample_prep_capabilities.md](#)
- Alignment QC package: [packages/methylalignmentqc/docs/USAGE.md](#)
- Extraction QC package: [packages/methylextractionqc/README.md](#)
- Historical design notes (cycle screening): [docs/plans/alignment-qc-screening.plan.md](#)

5 Package Reference (Individual Usage)

5.1 Purpose

Summarize each active package as a standalone tool, then map it to staged `methyl-validation` orchestration.

For the full SamplePrep story (Picard metrics, Parabricks JSON, guardrails, remediation, extraction QC), see [Chapter 3 — Sample Prep and Quality Control](#).

5.2 Package quick map

Package	CLI	Primary role	Typical standalone output
<code>methylalignmentqc</code>	<code>methyl-qc</code>	Parse alignment/dup metrics into normalized JSON	<code>alignment_qc/<sample>.json</code>
<code>methylcentroid</code>	<code>methyl-centroid</code>	Build cohort centroids from methylation files	centroid artifacts per group
<code>methyldetector</code>	<code>methyl-detector</code>	Discover/classify DMPs	<code>dmps-*-discovery.csv</code> , <code>result*.json</code>
<code>methylmapper</code>	<code>methyl-mapper</code>	Map DMP loci to genes/features	<code>all-gene_name-combined.csv</code>
<code>methylenricher</code>	<code>methyl-enricher</code>	Enrichment/module interpretation	<code>enrichment_merged.csv</code> , <code>modules_ranked.csv</code>
<code>methylclassifier</code>	<code>methyl-classifier</code>	Train production classifier (ECDF path)	<code>classifiers/*.pkl</code>
<code>methylpredictor</code>	<code>methyl-predictor</code>	Predict and report metrics/inference	<code>predictors/*/validation_metrics</code> or blind reports
<code>methylvalidation</code>	<code>methyl-validation</code>	Orchestrate MC/stability/freeze/model workflows	<code>monte_carlo_runs/*</code>
<code>methyl-disease-progression</code>	<code>methyl-disease-progression</code>	Cross-stage synthesis from mapper/enricher outputs	<code>progression/*.csv</code> , <code>summary.json</code>

5.3 Standalone examples

5.3.1 Alignment QC

```
source .venv/bin/activate
methyl-qc --metrics-root /data/metrics --output-dir /work/alignment_qc
```

5.3.2 Enrichment from mapper output

```
source .venv/bin/activate
methyl-enricher --input /work/project/mapper/pca_pca1/all-gene_name-combined.csv --modules --o
```

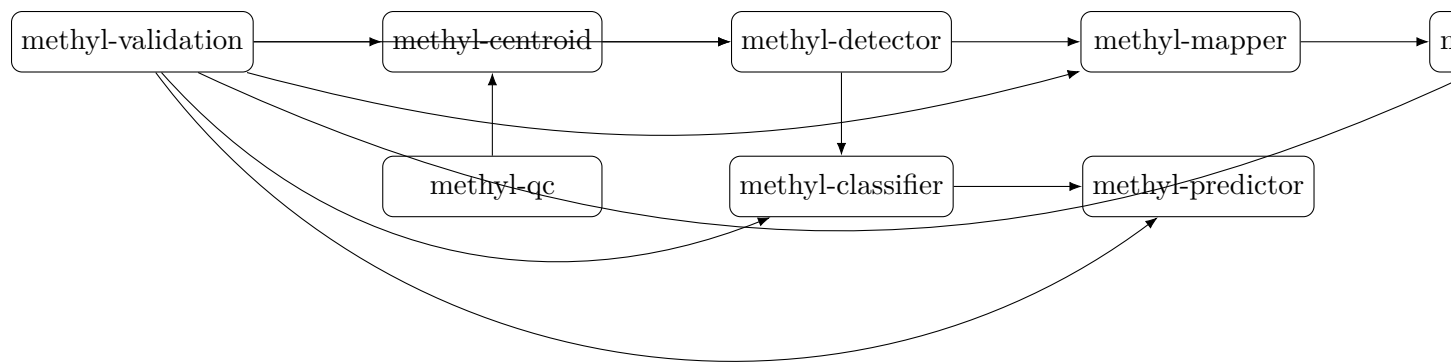
5.3.3 Blind prediction with frozen model

```
source .venv/bin/activate
methyl-predictor --project /work/project/monte_carlo_runs/production/project.json
```

5.4 Orchestrated equivalents in methyl-validation

- Stability stage runs centroid+detector per iteration.
- Freeze stage runs centroid->detector->mapper->enricher (and optional progression).
- Model stage runs:
 - ecdf: classifier->predictor
 - tabular_sklearn / generative_hybrid: in-process backend train/predict
- Post-model validation runs frozen-model descriptive holdouts.

5.5 Integration diagram



5.6 Do not do this

- Do not assume standalone package defaults match project-orchestrated overrides.
- Do not skip `project.json` artifact handoff checks when chaining packages manually.

5.7 See also

- `README.md`
- `packages/*/docs/USAGE.md`

Part II

Part II: Staged Solution Workflow

6 Stage: Stability

6.1 Purpose

Run repeated train/validation iterations to identify DMPs that recur across runs and materialize a stable production panel.

6.2 Entry command

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --
```

Optional strict FeatureCuts alignment:

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json \
  --stability --stability-featurecuts --stability-target-ba 0.95 --stability-min-selected-dmps
```

6.3 Required config keys

Under `step_config.validation`:

- `n_iterations`, `train_fraction`, `seed`
- `run_stability`
- `stability_dmp_freq`
- `stability_min_balanced_accuracy`
- `stability_gene_freq`
- optional adaptive stop controls:
 - `stability_early_stop_enabled`
 - `stability_min_iterations`
 - `stability_convergence_window`
 - `stability_convergence_jaccard`
 - `stability_convergence_max_size_delta`
 - `stability_convergence_patience`
- optional FeatureCuts controls:

- `stability_featurecuts_enabled`
- `stability_target_balanced_accuracy`
- `stability_min_selected_dmps`

6.4 Composable workflow profiles

DomainProgram workflows support selectable paths via context profiles in `workflow_engine/domain/profiles/`. Use `methy1-validation run-workflow` or `methy1-workflow-run` with `--context-file`.

Gene stability without DMP FeatureCuts (discovery DMPs → mapper → enricher → stability on enricher genes):

- Profile: `gene_enricher_stability.profile.json`
- Program: `mc_gene_enricher_stability.program.json`
- Set `stability_featurecuts_enabled: false`, `stability_gene_featurecuts_enabled: false`, `stability_gene_freq` as needed
- Mapper uses `dmps-*-discovery.csv`; enricher applies disease filters from `step_config.enricher`

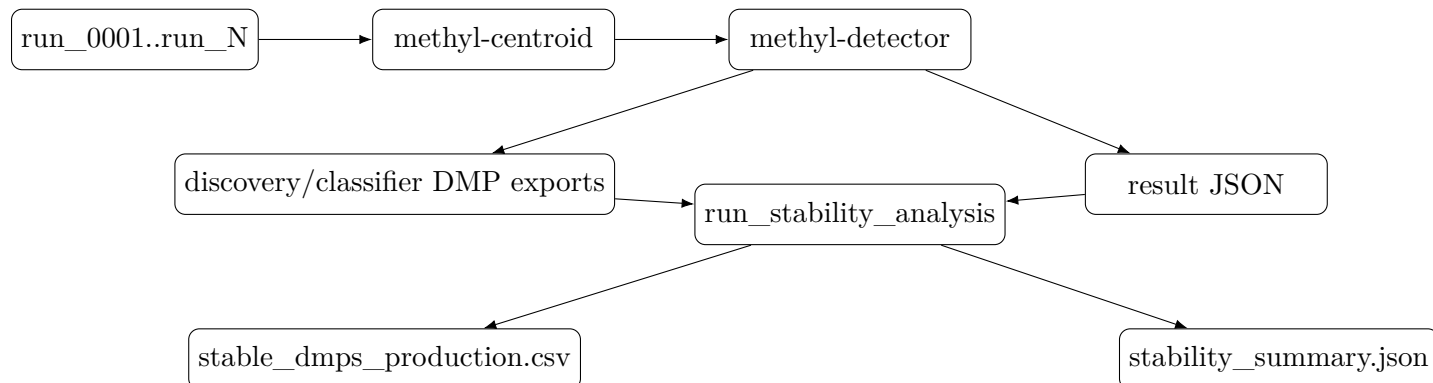
DMP panel stability (split detector + `pipeline.dmp_select`): profile `dmp_panel_stability` or `full_biomarker_gene_fc`.

Discovery DMPs → broad mapper → gene FeatureCuts (stable gene panel for model): profile `discovery_gene_featurecuts` with `pca1_5_mc_stability.program.json`. Mapper uses `dmps-*-discovery.csv`; `pipeline.dmp_select` still runs so `pipeline.gene_select` has classifier DMP exports for validation. Biomarker/PPI pre-filter is off by default (largest gene pool); enable with `"runBiomarkerFilter": true` in context.

Biomarker / PPI pre-filter (`validation.biomarker_filter`): optional via `${runBiomarkerFilter}`; modes `ppi_only`, `disease_only`, `disease_and_ppi` under `step_config.validation` or `gene_selection`. Can run on mapper gene pools without gene FeatureCuts.

See `docs/domain_program_language.md` for the full profile matrix and IF branch flags (`runDmpSelection`, `runGeneFeaturecuts`, ...).

6.5 Process model



6.6 Expected outputs

- `<mc_root>/stability/stable_dmps_production.csv`
- `<mc_root>/stability/stability_summary.json`
- `<mc_root>/all_metrics.csv`
- `<mc_root>/metrics_summary.json`
- `<mc_root>/baseline_manifest.json`

Where `<mc_root>` is: `<output_base>/<project_name>/monte_carlo_runs/`

6.7 Success checks

- `stable_dmps_production.csv` exists and is non-empty.
- `all_metrics.csv` has successful rows near expected run count.
- `stability_summary.json` reports configured thresholds.
- if adaptive stop is enabled, `stability_summary.json` includes `early_stopping` diagnostics (triggered, checkpoint history, and stop iteration when applicable).

6.8 Do not do this

- Do not proceed to freeze when stable panel is missing or empty.
- Do not interpret blind outputs as stability quality evidence.

6.9 Recovery

- Resume interrupted run:

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --
```

- If panel is too small: lower `stability_dmp_freq` or increase `n_iterations`.
- If run quality is too low: tune detector and/or adjust `stability_min_balanced_accuracy`.

6.10 See also

- `docs/theory/chapters/12-two-workflows.qmd`
- `docs/theory/chapters/15-model-creation-and-validation.qmd`
- If many MC iterations are split across several hosts, use shared `output_base` and the distributed queue flow: [Distributed MethylValidation on shared storage](#) (12).

7 Stage: Freeze

7.1 Purpose

Create frozen production artifacts using the stable panel and generate biological interpretation outputs (mapper/enricher, optional progression).

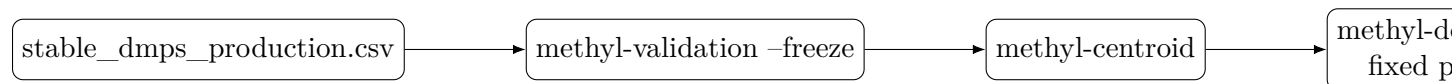
7.2 Entry command

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --
```

7.3 Prerequisites

- Stability stage completed.
- Stable panel available at:
 - configured `step_config.validation.freeze_stable_dmp_csv`, or
 - default `<mc_root>/stability/stable_dmps_production.csv`.

7.4 Process model



7.5 Expected outputs

- `<mc_root>/production/project.json`
- `<mc_root>/production/production_summary.json`
- `<mc_root>/production/model_bundle/mapper_dmp_annotations.csv` (when mapped-family feature bundles are prepared)
- mapper outputs under `production/mapper/...`
- enricher outputs under `production/enricher/...`
- optional progression outputs under `production/progression/...`

7.6 Success checks

- `production/project.json` exists and includes fixed panel usage for detector.
- expected mapper and enricher files are present.
- stage-specific logs show successful return codes.
- for non-dmp observed-hybrid families, `production_summary.json` includes `mapper_annotation_cache` metadata.

7.7 Recommended post-freeze gate

Run biological readiness before model training:

```
source .venv/bin/activate
methyl-validation biological-readiness /work/prostate-cancer/Healthy_vs_PCa1-4-CG
```

This chained gate verifies:

- enricher completeness (`verify-complete`),
- strict progression input completeness (`--strict-missing`),
- stability/freeze/progression readiness verdict.

To include advisory Grok commentary in the readiness section:

```
methyl-validation biological-readiness /work/prostate-cancer/Healthy_vs_PCa1-4-CG --grok-review
```

Without `--grok-review`, the chain keeps Grok disabled by default and still enforces deterministic checks.

7.8 Do not do this

- Do not run freeze before stability panel exists.
- Do not delete `production/project.json`; later model and predictor stages depend on it.

7.9 Recovery

- If freeze fails because stable panel missing, rerun stability or set `freeze_stable_dmp_csv`.
- If path mismatch occurs across machines, use `path_remap` or `--path-remap`.

7.10 See also

- `packages/methylvalidation/docs/USAGE.md`
- `docs/theory/chapters/14-user-guide.qmd`

8 Stage: Model

8.1 Purpose

Train and select final production model artifacts after freeze and biological-readiness checks.

8.2 Common execution patterns

8.2.1 Pattern A: Direct production model build

```
source .venv/bin/activate
methyl-validation biological-readiness /work/prostate-cancer/Healthy_vs_PCa1-4-CG
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --r
```

8.2.2 Pattern B: Backend model-MC comparison then select best

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --r
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json \
  --select-best-model --model-mc-all --selection-metric balanced_accuracy --selection-stat med
```

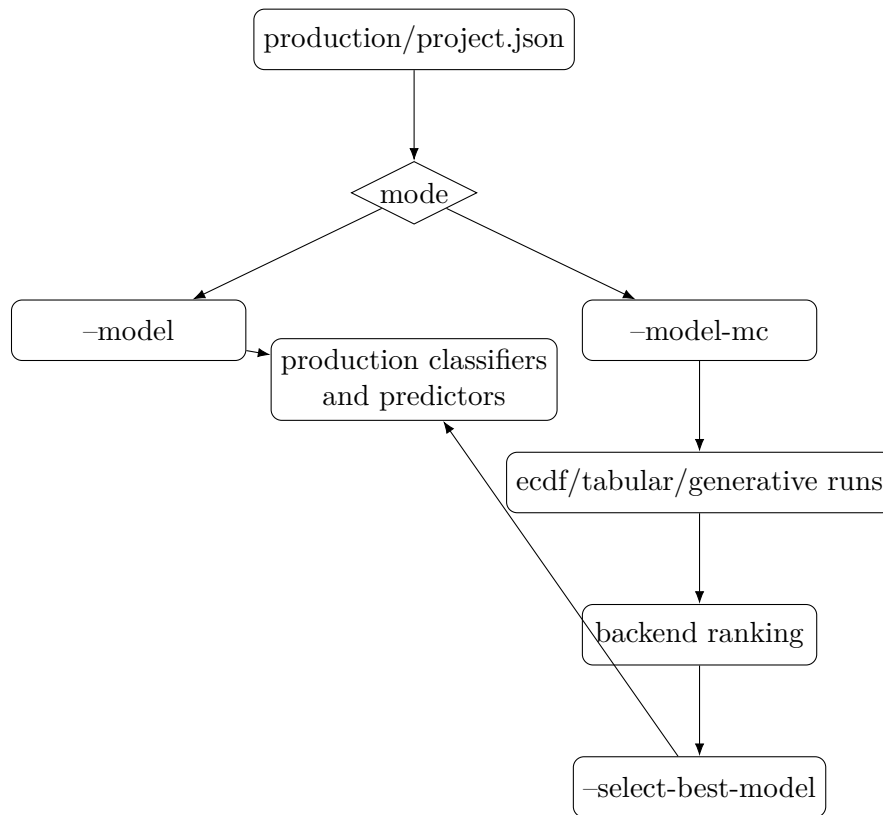
8.3 Backend behavior

- `ecdf`: `methyl-classifier` -> `methyl-predictor`
- `tabular_sklearn`: `in-process bundle` -> `train` -> `predict`
- `generative_hybrid`: `in-process bundle` -> `train` -> `predict`

For `ecdf` + observed-hybrid mapped families (`feature_mode=observed_hybrid`, `feature_family_set!=dmp_score`) the runtime path is aggregated ECDF OvR package `train/predict` and skips ECDF second-stage refinement.

Configured by `step_config.validation.backend_profiles` (plus optional CLI backend override).

8.4 Process model



8.5 Required config keys

- `backend_profiles`
- backend-specific `params` under each profile (for example `feature_mode`, `ecdf_second_stage_enabled`, `tabular_methods`, `generative_latent_dim`)
- observed-hybrid family contract (when used):
 - `feature_family_set`
 - mapper annotation cache from freeze (`production/model_bundle/mapper_dmp_annotations.csv`)
- tabular controls:
 - `tabular_methods`
 - `tabular_method_selection_metric`
 - optional `tabular_max_dmps` (null/0 keeps all stable loci)
- generative controls:
 - `generative_latent_dim`, `generative_epochs`, and related keys
- optional gate:
 - `require_biological_review_for_model`
 - `biological_review_confirmed`

Legacy flat backend keys are rejected. Migrate older configs with:

```
methy1-validation-migrate-backend-config /path/to/project.json --in-place
```

8.6 Expected outputs

- `<mc_root>/production/classifiers/...`
- `<mc_root>/production/predictors/...`
- aggregated ECDF mode writes:
 - `<mc_root>/production/classifiers/ecdf_aggregated_ovr.pkl`
 - `<mc_root>/production/classifiers/ecdf_aggregated_ovr.meta.json`
- if model-MC:
 - `<mc_root>/model_mc/shared/run_XXXX/...`
 - `<mc_root>/model_mc/<backend>/all_metrics.csv`
 - `<mc_root>/model_mc/backend_ranking.csv`
 - `<mc_root>/production/selected_backend.json` (after selection)

When model-MC reuses primary split artifacts, shared/backend run roots may link centroid/detector directories instead of recomputing them.

8.7 Success checks

- final classifier/predictor artifacts exist in production root.
- selected backend metadata exists when using selection workflow.
- metrics summaries include expected selection metric fields.
- the last biological-readiness report is `go/go_with_risks` and reviewed by humans.

8.8 Do not do this

- Do not run model before freeze.
- Do not skip biological readiness review when production artifacts changed.
- Do not claim backend superiority without model-MC summaries.

8.9 See also

- `docs/theory/chapters/15-model-creation-and-validation.qmd`
- `packages/methy1validation/docs/ROLLOUT.md`

9 Stage: Post-model Validation

9.1 Purpose

Measure descriptive performance distributions using frozen production artifacts without retraining.

9.2 Entry command

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --]
```

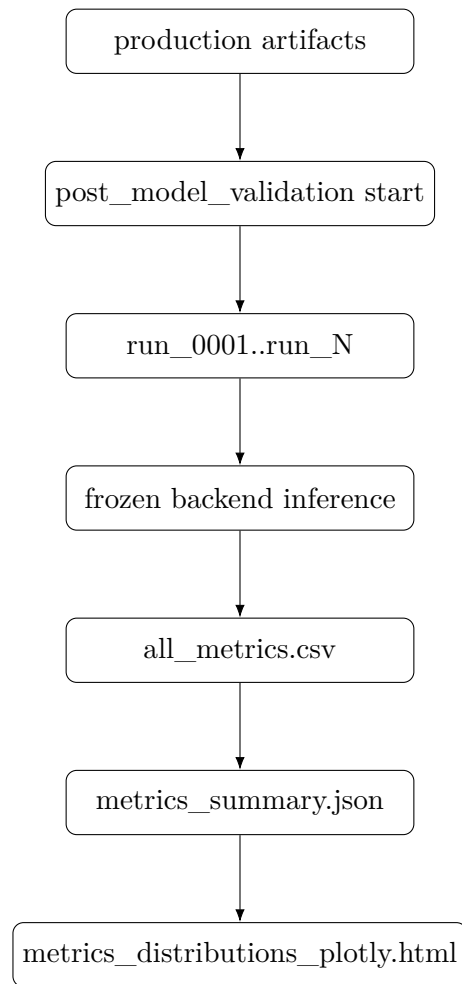
Optional backend override:

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json \
  --post-model-validation --post-model-backend ecdf
```

9.3 Prerequisites

- Freeze/model artifacts already built in production root.
- Valid production/project.json.

9.4 Process model



9.5 Expected outputs

- `<mc_root>/post_model_validation/run_XXXX/...`
- `<mc_root>/post_model_validation/all_metrics.csv`
- `<mc_root>/post_model_validation/metrics_summary.json`
- `<mc_root>/post_model_validation/metrics_distributions_plotly.html`

9.6 Success checks

- run directories created under `post_model_validation`.
- metrics summary includes key metrics (`balanced_accuracy`, `macro_f1`, plus available calibration metrics).
- distribution plot generated.

9.7 Do not do this

- Do not treat post-model validation as model training.
- Do not compare candidate and baseline manually without fixed thresholds; use rollout comparison when promoting.

9.8 Recovery

- Resume interrupted post-model validation:

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json \
  --post-model-validation --resume
```

9.9 See also

- `packages/methylvalidation/docs/USAGE.md`
- `packages/methylvalidation/docs/ROLLOUT.md`

10 Stage: Blind Prediction

10.1 Purpose

Run inference on newly arrived unlabeled samples using frozen model artifacts.

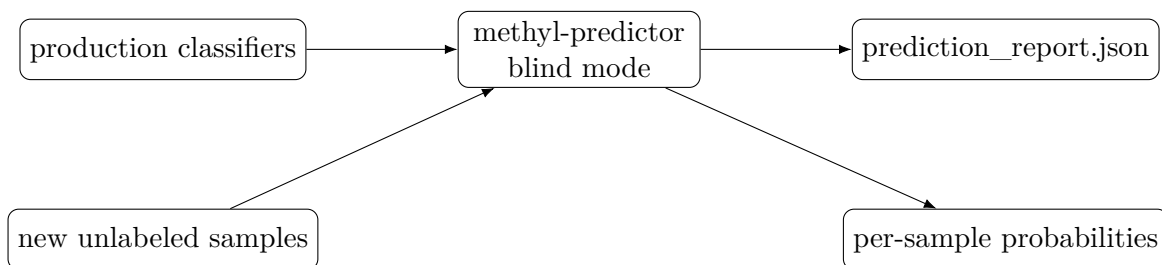
10.2 Important boundary

Blind-only prediction is a standalone `methyl-predictor` operation. It is intentionally outside `methyl-validation` accuracy workflows.

10.3 Entry command (standalone)

```
source .venv/bin/activate
methyl-predictor --project /work/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/product.
```

10.4 Process model



10.5 Required config pattern

`step_config.predictor` should reference blind/test-blind inputs and model paths resolved from frozen production project.

10.6 Expected outputs

- blind prediction report JSON
- per-sample class probabilities and predicted subgroup labels

10.7 Success checks

- predictor completes without missing model-path errors.
- expected blind report file exists.
- per-sample prediction rows match input sample count.

10.8 Do not do this

- Do not use blind-only prediction results as labeled validation accuracy claims.
- Do not run blind-only predictor settings through `methy1-validation` MC paths.

10.9 Recovery

- If model path resolution fails, confirm `production/project.json` and classifier artifacts exist.
- If sample-id join fails, verify sample path naming and predictor input lists.

10.10 See also

- `packages/methylpredictor/docs/USAGE.md`
- `docs/theory/chapters/14-user-guide.qmd`

Part III

Part III: Operations and Reliability

11 Artifacts and QA Checks

11.1 Purpose

Provide a single checklist for expected files at each stage and quick quality gates before proceeding.

11.2 Workflow action observability

Each workflow ACTION executed by `methy1-worker` submits a typed `output_json` to the database (validated against `schemas/tasks/*.json`). Outputs include timing (`duration_ms`), branch code (`result_code`), and paths to artifacts on shared storage.

On `/work`, look for:

- **Per-action manifests:** `{output_dir}/.action_results/{action_name}.{run_key}.json` — full typed result written by CLIs (e.g. `pipeline.dmp_select`, `pipeline.mapper`) or reconstructed by the worker collector.
- **Sample prep timeline:** `{sampleDir}/{sampleId}.sample_prep_log.jsonl` — append-only JSONL audit for download, align, QC, extract, archive steps.
- **Validation / MC timeline:** `{monteCarloRunsRoot}/action_run_log.jsonl` — append-only log for validation-category actions (plan, stability, model_mc, etc.).

Regenerate JSON Schema artifacts after model changes:

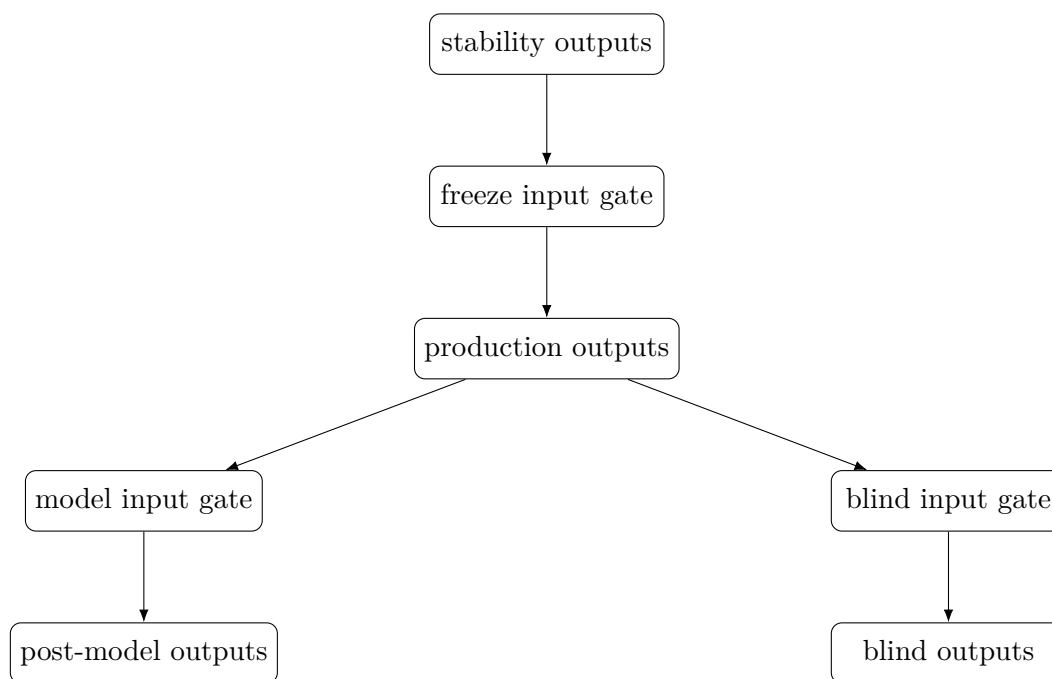
```
source .venv/bin/activate
methy1-export-task-schemas
methy1-export-task-schemas --check # CI drift gate
```

11.3 Artifact map

Stage	Required artifacts	Gate to proceed
Sample prep	<code>{sample_id}.json</code> , <code>alignment_qc/{sample_id}.json</code> , <code>{sample_id}.extraction_qc.json</code> , <code>{chr}-CG.h5</code>	<code>guardrails.overall_pass</code> true for alignment and extraction QC
Stability	<code>stability/stable_dmps_production/</code> <code>stability_summary.json</code> , <code>all_metrics.csv</code>	Stable panel exists and is non-empty

Stage	Required artifacts	Gate to proceed
Freeze	<code>production/project.json</code> , mapper/enricher outputs	Production project exists; fixed-panel detector run completed
Model	<code>production/classifiers/*</code> , <code>production/classifiers/training_metrics.json</code> (all model backends), <code>production/predictors/*</code>	Final classifier/predictor artifacts present
Post-model validation	<code>post_model_validation/all_model_validation metrics_summary.json</code> , plotly report	Distribution summaries generated
Blind prediction	blind prediction report + per-sample probabilities	All input blind samples receive predictions

11.4 Handoff integrity checks



11.5 QA checklist per stage

11.5.1 Stability

- Iteration count in metrics roughly matches configured `n_iterations`.
- Stable panel size is consistent with selected thresholds.

11.5.2 Freeze

- `production/project.json` references fixed panel detector mode.
- Mapper/enricher outputs exist for each expected comparison group.

11.5.3 Model

- Backend used is recorded and consistent with config.
- If selection used, ranking artifacts exist and chosen backend is traceable.

11.5.4 Post-model validation

- Metrics summary includes required decision metrics.
- Distribution artifacts generated and readable.

11.5.5 Blind prediction

- Prediction outputs include all requested samples.
- Operational report does not include false accuracy claims.

11.6 Escalation triggers

- Empty stable panel.
- Missing production project.
- No model artifacts after model stage.
- Repeated run failures in same step across resumed runs.

11.7 See also

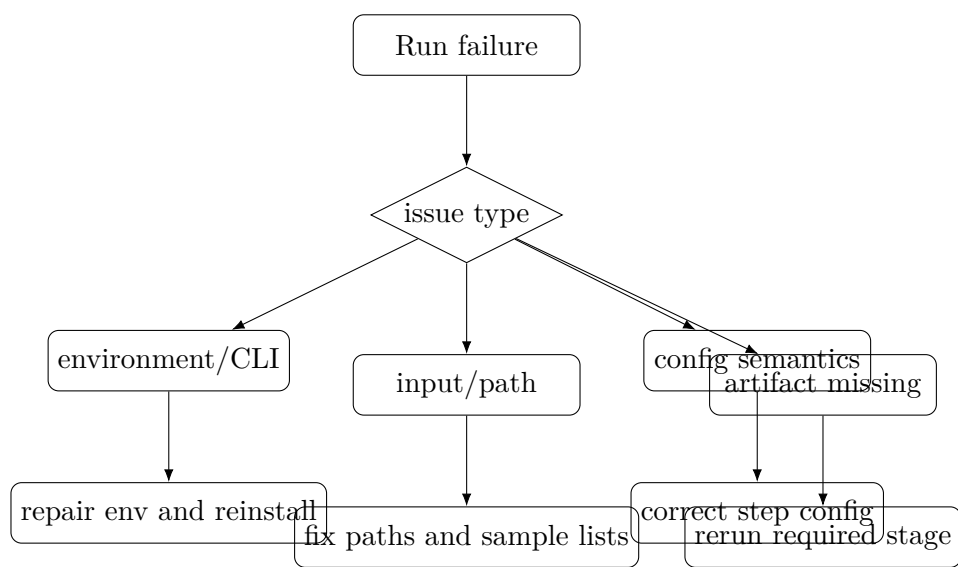
- `packages/methylvalidation/docs/USAGE.md`
- `packages/methylvalidation/docs/IMPLEMENTATION.md`

12 Troubleshooting and Recovery

12.1 Purpose

Standardize response for common stage failures and reduce ad-hoc debugging.

12.2 Failure routing



12.3 Common problems

Symptom	Likely cause	Recovery
command not found: methyl-validation	.venv inactive or install missing	Activate .venv, rerun scripts/install_all.sh
freeze fails for missing stable panel	stability not run or wrong path	Run --stability or set freeze_stable_dmp_csv
model fails due to biological review gate	require_biological_review_for_xenome and not confirmed	set model=true biological_review_confirmed=true after review
path file-not-found across hosts	environment path differences	use path_remap/--path-remap OLD=NEW

Symptom	Likely cause	Recovery
blind predictor rejected in validation mode	blind settings passed to <code>methyl-validation</code>	run blind mode with standalone <code>methyl-predictor</code>

12.4 Safe recovery sequence

1. Validate environment and CLI availability.
2. Confirm previous-stage artifacts exist.
3. Re-run failed stage only.
4. Use resume mode where supported.
5. Re-check QA gates before advancing.

12.5 Resume examples

```
source .venv/bin/activate
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --resume
methyl-validation --project /work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json --resume
```

12.6 Operational guardrails

- Keep original run directories for traceability.
- Prefer additive reruns over manual artifact edits.
- Record config overrides used in reruns.

12.7 See also

- `docs/code_first_discovery_report.md`
- `docs/config_parameter_matrix.md`

13 Command Cookbook

13.1 Purpose

Provide copy-ready command sequences for the full solution lifecycle.

13.2 1) Full staged run (strict profile example)

```
source .venv/bin/activate
PROJECT="/work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"

methyl-validation --project "$PROJECT" --stability
methyl-validation --project "$PROJECT" --freeze
methyl-validation --project "$PROJECT" --model
methyl-validation --project "$PROJECT" --post-model-validation
```

13.3 2) Backend comparison and selection

```
source .venv/bin/activate
PROJECT="/work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"

methyl-validation --project "$PROJECT" --model-mc --model-mc-all
methyl-validation --project "$PROJECT" --select-best-model --model-mc-all \
  --selection-metric balanced_accuracy --selection-stat median
```

13.4 3) Rollout comparison report

```
source .venv/bin/activate
PROJECT="/work/prostate-cancer/configs/project_Healthy_vs_PCa1-4-CG.json"
BASE="/work/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/model_mc/ecdf/metrics_summary"
CAND="/work/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/model_mc/tabular_sklearn/metrics_summary"

methyl-validation --project "$PROJECT" --rollout-compare \
  --baseline-summary "$BASE" --candidate-summary "$CAND"
```

13.5 4) Standalone enricher run from mapper output

```
source .venv/bin/activate
methyl-enricher \
  --input /work/prostate-cancer/Healthy_vs_PCa1-4-CG/mapper/all/pca_pca1/all-gene_name-combined
  --modules \
  --library-preset cancer-extended \
  --outdir /work/prostate-cancer/Healthy_vs_PCa1-4-CG/enricher/all/pca_pca1
```

13.6 5) Standalone blind prediction

```
source .venv/bin/activate
methyl-predictor --project /work/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/product.
```

13.7 6) Manual rendering commands (this manual)

```
quarto render docs/user-manual --to html
quarto render docs/user-manual --to pdf
```

13.8 7) Distributed queue (one task)

Full workflow: [Distributed MethylValidation on shared storage](#) (12). One worker invocation:

```
source .venv/bin/activate
methyl-validation run-task --task /work/.../monte_carlo_runs/queue/tasks/run_0001.json
```

13.9 8) Hyperparameter search grid (example)

After `plan-runs` / full stability is understood, use the grid driver. Details: [Optional hyperparameter search](#) (13).

```
source .venv/bin/activate
methyl-hyperparam-search \
  --config /work/experiments/my_mc_config.json \
  --work-dir /work/experiments/hp_run1 \
  --grid '{"stability_dmp_freq": [0.6, 0.7]}' \
  -- --stability
```


13.10 Execution notes

- Always activate `.venv` before `methyl-*` commands.
- Use absolute project paths in production scripts.
- Keep command logs with timestamp and git commit hash for audit.

14 Distributed MethylValidation on shared storage

14.1 Purpose

Run discovery Monte Carlo (and related queue-backed work) on **several machines** when a single host cannot finish `n_iterations` in acceptable wall time. Every worker must read and write the **same** `output_base` tree (commonly shared NFS under `/work/...`).

14.2 Prerequisites

- One `MonteCarloConfig` (dedicated `mc.json` or a project with `step_config.validation`) with `output_base` on shared storage
- The same virtual environment and `methyl-*` CLIs on each worker
- R/W access to: `output_base/<project_name>/monte_carlo_runs/` and the `queue/` subtree created by planning

14.3 Layout (under `monte_carlo_runs/`)

Path	Role
<code>queue/mc_config.json</code>	Snapshotted config for workers
<code>queue/plan_runs.json</code>	Planned runs and task JSON paths
<code>queue/tasks/run_####.json</code>	One task = one <code>run-task</code> invocation
<code>queue/queue_manifest.jsonl</code> / <code>queue/commands.sh</code>	Optional export for a job scheduler
<code>run_####/queue_task_status.json</code>	Worker completion status
<code>run_####/queue_local_step_timings.json</code>	Merged by <code>aggregate-results</code> into <code>step_timings.csv</code>

Full path reference: `packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md`.

14.4 Command sequence

1) **Plan** (one process; creates run dirs and task JSON, no heavy steps):

```
source .venv/bin/activate
methyl-validation plan-runs --config /work/experiments/my_mc_config.json --overwrite
# or: methyl-validation plan-runs --project /work/prostate-cancer/configs/project_Healthy_vs_L
```

`--overwrite` only replaces `queue/tasks/` and regenerates the plan; it does **not** remove existing `run_####` output directories. To delete all `run_####` trees before planning, add `--wipe-runs` (irreversible). Without `--overwrite`, completed runs (see `queue_task_status.json` + per-run task under `queue/tasks/`) are not written over; you can add iterations and only the new `run_####` work is planned. `run-task` skips pipeline work for runs already marked completed unless you pass `--force`.

2) **Export** (optional; for a central queue broker):

```
methyl-validation export-queue --config /work/experiments/my_mc_config.json
```

3) **Workers** (one job per task; repeat until the queue is empty):

```
methyl-validation run-task --task /work/prostate-cancer/Healthy_vs_PCa1-4-CG/monte_carlo_runs/
```

4) **Aggregate** (after successful tasks, or to refresh from a partial set):

```
methyl-validation aggregate-results --config /work/experiments/my_mc_config.json
# add --stability to run the same DMP/gene stability pass as monolithic methyl-validation --st
```

If adaptive stop settings are enabled for monolithic stability runs, remember queue workers do not coordinate global early termination on their own. Keep `stability_early_stop_enabled=false` for queue-first executions unless your scheduler supports coordinator-side cancellation logic.

Legacy: a single process may still run `methyl-validation --stability` without subcommands; behavior is unchanged.

14.5 Example paths (/work pattern)

With `output_base` pointing at shared storage, `plan-runs / export-queue` resolve `monte_carlo_runs` the same way as the monolithic CLI: `output_base / project_name / monte_carlo_runs /`.

14.6 Recovery

- Re-run a failed `run-task` for the same `--task` path.
- `queue_task_status.json` records failed vs completed.
- `aggregate-results` skips failed runs (when status files are present) and includes runs with computable per-run metrics.

14.7 See also

- `packages/methylvalidation/docs/DISTRIBUTED_QUEUE.md` (canonical detail)
- `docs/theory/chapters/15-model-creation-and-validation.qmd` (section *Distributed Monte Carlo on shared storage*)
- [Optional hyperparameter search](#) (search trials can use the same shared storage)

15 Optional hyperparameter search

15.1 When to use

After a baseline `methyl-validation` path is understood, use a controlled **outer loop** to compare candidate settings—typically **Tier A** knobs in `step_config.validation` / `MonteCarloConfig` (e.g. `stability_dmp_freq`, `n_iterations` as budget, `stability_min_balanced_accuracy`). Tuning detector and model backends (Tier B/C) is a separate, often **nested** exercise; see the theory chapter on multi-stage search tradeoffs in `docs/theory/chapters/15-model-creation-and-validation.qmd` (*Optional pipeline hyperparameter search*).

15.2 What it is

- A **read-only** scalar **objective** (J) is computed from artifacts under `monte_carlo_runs/` (mainly `metrics_summary.json`, optional `stability/stability_summary.json`).
- The `methyl-hyperparam-search` entry point is a small **grid** driver: for each grid point it writes `work_dir/trial_NNNN/mc_config.json` with a merged `MonteCarloConfig` and a distinct `output_base`, runs `methyl-validation` (e.g. with `--stability`), then scores the run.
- It does **not** use `sklearn.model_selection` CV: each evaluation is a subprocess and file read.

Formula and weights: `packages/methylvalidation/docs/HYPERPARAMETER_SEARCH.md` (full (J), `ObjectiveWeights`, constraint semantics).

15.3 Command example

Use a dedicated MC config and a work directory on shared storage if many trials:

```
source .venv/bin/activate
methyl-hyperparam-search \
  --config /work/experiments/my_mc_config.json \
  --work-dir /work/experiments/hp_run1 \
  --grid '{"stability_dmp_freq": [0.6, 0.7], "stability_min_balanced_accuracy": [null, 0.5]}' \
  -- \
  --stability
```

- JSON `null` in `--grid` omits that field so the base config value is kept.
- Results: `work-dir/search_summary.json` (per-trial rows and best feasible trial, if any).
- `--weights-json` can point to a file for `ObjectiveWeights`.

- `--baseline-summary /path/to/metrics_summary.json` enforces **rollout-style** feasibility vs a baseline (same idea as `methyl-validation --rollout-compare`); see `packages/methylvalidation/docs/ROLLOUT.md` and `packages/methylvalidation/docs/HYPERPARAMETER`
- `--dry-run` only writes trial configs and paths; no pipeline subprocesses.

15.4 Pitfalls

- **Discovery vs calibration:** settings that optimize stability panel size may not optimize NLL, Brier, or ECE; prefer a **two-stage** strategy or tight constraints, not a single ad hoc weight mix.
- **n_iterations:** treat primarily as **compute budget** to stabilize medians, not a dimension to grid-search widely without cost awareness.

15.5 See also

- `packages/methylvalidation/docs/HYPERPARAMETER_SEARCH.md`
- `packages/methylvalidation/docs/ROLLOUT.md` (promotion / dual-run thinking)
- `docs/theory/chapters/15-model-creation-and-validation.qmd` (*Optional pipeline hyperparameter search*)
- [Distributed MethylValidation on shared storage](#) (for parallel execution of a single config's iterations)